MANGROVE GUARDIAN

An AI Agent for Satellite-Based Monitoring and Conservation of Mangrove Forests

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Abstract: Mangroves provide coastal protection, store carbon, support biodiversity, and filter water. Despite these benefits, they are increasingly threatened by climate change, pollution, land-use change, cyclones, salinity stress, and other anthropogenic pressures. The Sundarbans, shared by India and Bangladesh, form the world's largest contiguous mangrove forest and are an ecological treasure of global significance. Existing monitoring approaches commonly rely on field surveys, periodic remote-sensing analysis, or standalone machine-learning classification. Although useful, these approaches do not always provide continuous, integrated decision support. This paper proposes Mangrove Guardian, an AI-assisted framework that combines Sentinel-2 vegetation observations with Sentinel-5P atmospheric measurements. Sentinel-2 surface-reflectance imagery is used to calculate the Normalized Difference Vegetation Index (NDVI), while Sentinel-5P/TROPOMI provides carbon monoxide (CO) column measurements as an indicator of atmospheric conditions. Temporally aligned NDVI and CO features are organized into a structured dataset for Random Forest-based prediction of vegetation condition or degradation risk. A decision engine interprets predictions, detects unusual changes, assigns risk levels, recommends conservation actions, and generates alerts for an interactive dashboard. The proposed architecture is modular and repeatable, with a pathway from a research prototype to a broader environmental decision-support platform. Its main contribution is the integration of vegetation monitoring, atmospheric context, predictive analytics, and agent-oriented conservation support.

Keywords: Mangrove monitoring; Sundarbans; Sentinel-2; Sentinel-5P; NDVI; carbon monoxide; Google Earth Engine; environmental decision support; AI agents; Random Forest.

1. Introduction

Mangrove forests occur at the dynamic boundary between land and sea and provide valuable ecosystem services. They reduce coastal erosion and wave energy, help limit storm-surge impacts, support fisheries and biodiversity, filter sediments and pollutants, and sequester substantial amounts of carbon. The Food and Agriculture Organization (FAO) estimates that mangroves cover approximately 14.8 million hectares worldwide. Despite their ecological and socioeconomic importance, these ecosystems remain threatened by environmental pressures and degradation [1], [2].

The Sundarbans are a particularly important example. Located in the Ganges–Brahmaputra–Meghna delta and shared by India and Bangladesh, they form the world's largest contiguous mangrove forest. The ecosystem includes tidal waterways, mudflats, and salt-tolerant forests. Monitoring is essential because the region is exposed to cyclones, human pressures, changing hydrology, and climate-related stress [3], [4].

Traditional field surveys can provide detailed ground observations, but they are costly, time-consuming, and difficult to conduct across extensive, inaccessible tidal areas. Satellite remote sensing offers a scalable alternative. Sentinel-2 provides multispectral observations with a nominal five-day revisit cycle, while Google Earth Engine (GEE) supports large-scale cloud processing of geospatial data and rapid development of monitoring workflows [5], [6]. Recent studies have demonstrated the value of Sentinel imagery and GEE for analyzing vegetation dynamics, forest fragmentation, and land-use change in the Sundarbans [7]–[10].

However, a gap remains between producing environmental indicators and delivering actionable conservation intelligence. Many systems focus on map visualization or land-cover classification without integrating multiple environmental signals, forecasting conditions, assessing risk, recommending actions, and issuing alerts through a single workflow. Mangrove Guardian is proposed to help address this gap.

2. Related Work

Recent literature demonstrates strong interest in satellite-based mangrove monitoring, particularly through GEE. Common approaches combine Sentinel or Landsat imagery, vegetation indices such as NDVI and EVI, and machine-learning methods including Random Forest, support vector machines, and neural networks. Kanjin and Alam [7] examined long-term land-cover, NDVI, and land-surface-temperature changes in the Sundarbans, illustrating the value of remote sensing for detecting ecosystem change. Roy et al. [8] investigated forest dynamics using high-resolution Sentinel-2 imagery and GEE.

Studies of the Bangladesh Sundarbans have also examined land-cover classification and spatial fragmentation during 2017–2023. Research in the Sundarban Biosphere Reserve has combined Sentinel-1, Sentinel-2, and GEE to investigate degradation and vegetation decline associated with extreme climatic events [9]. A long-term Landsat time-series study published in 2025 reported vegetation changes and browning signals in the Sundarbans, highlighting the importance of temporal trends and environmental drivers [10].

These studies support continuous satellite observation, but they do not necessarily extend to an autonomous alert-and-recommendation workflow. Niceline et al. [11] proposed a GEE-based mangrove-degradation framework beyond the Sundarbans. Their work on Pichavaram used variables including Sentinel-2 and Landsat data, NDVI, NBR, topography, salinity, tidal dynamics, and proximity measures with Random Forest classification. This demonstrates the value of multifactor modelling and motivates future extensions of Mangrove Guardian beyond NDVI and CO.

Global platforms such as Global Mangrove Watch (GMW) provide important information on mangrove extent, change, and alerts at broad geographic scales [12]. These resources make a significant contribution to conservation, while project-specific systems can complement them with localized predictive analysis, configurable decision logic, and explainable workflows. The literature therefore points to a research opportunity: integrating vegetation condition, atmospheric context, predictive machine learning, anomaly detection, and decision support in one modular monitoring pipeline.

3. Research Gap and Novelty

The principal research gap is not the absence of satellite monitoring, but the limited integration of monitoring outputs into a complete decision-support pipeline. Existing research commonly addresses vegetation mapping, land-cover classification, fragmentation, carbon estimation, or long-term trend analysis. Fewer systems combine these outputs with a decision layer that continuously evaluates changes and recommends conservation actions.

The proposed contributions of Mangrove Guardian are:

- Integrating complementary environmental observations: vegetation condition measured through NDVI and atmospheric CO signals.
- Reducing local storage and processing requirements through cloud-native processing in Google Earth Engine.
- Establishing a clear Random Forest baseline for machine-learning prediction and benchmarking.
- Converting model outputs into interpretable vegetation-condition classes (healthy, moderate, or degraded) and associated risk levels.
- Extending beyond visualization to provide recommendations and alerts.
- Providing a modular, scalable agent architecture for data collection, prediction, decision-making, recommendations, alerts, and reporting.

4. Methodology

4.1 Study Area

Study area: Sundarbans Mangrove Ecosystem, India–Bangladesh

The transboundary Sundarbans mangrove ecosystem is selected because of its global ecological significance, sensitivity to environmental change, and availability of long-term satellite observations. The study will use a clearly documented area-of-interest (AOI) polygon to define the spatial boundary. The final boundary and analysis period should be specified explicitly for reproducibility.

4.2 Sentinel-2 Processing and NDVI

Sentinel-2 processing will use the harmonized Level-2A surface-reflectance collection in GEE. Images will be filtered by the AOI, observation period, and cloud conditions. NDVI is calculated from the near-infrared (B8) and red (B4) bands using NDVI = (NIR − Red) / (NIR + Red). Higher positive values generally indicate denser or more photosynthetically active vegetation, while lower values may indicate sparse vegetation, stress, water, or non-vegetated surfaces. Interpretation should account for seasonality and the characteristics of the study area rather than relying on universal thresholds.

4.3 Sentinel-5P Processing

The GEE collection COPERNICUS/S5P/OFFL/L3_CO provides carbon monoxide observations from the TROPOMI instrument. CO is an atmospheric trace gas associated with combustion, biomass burning, and other processes [13]. In this study, CO is treated as an environmental-context variable, not as a direct causal measure of mangrove degradation. Differences in spatial resolution between Sentinel-2 and Sentinel-5P must be handled explicitly during spatial aggregation and temporal alignment.

4.4 Temporal Aggregation

Monthly summaries or composites will be generated to reduce noise and support comparison across sensors. Median or mean NDVI and CO values will be calculated for each period. Additional features may include rolling averages, month, day of year, year, changes in NDVI over defined intervals, and CO anomalies.

4.5 Data Preparation

Missing observations, cloud contamination, and inconsistent timestamps will be examined. Records will be standardized and documented before being divided into training and test sets. Where possible, a time-aware split will be used to reduce overly optimistic evaluation caused by temporal data leakage.

4.6 Machine Learning

Random Forest is selected as the initial baseline because it can model nonlinear relationships and interactions in noisy tabular data and provides feature-importance estimates. Hyperparameters, including the number of trees, maximum depth, minimum samples per leaf, and class weights, will be tuned using validation data.

4.7 Labeling Strategy and Evaluation

For categorical health-state prediction, labels should be based on documented ecological thresholds, independent land-cover or degradation assessments, expert interpretation, or validated change events. Labels derived solely from NDVI should not be used to claim that a model independently predicts the same NDVI-derived classes, as this would create circular evaluation. If reliable labels are unavailable, the initial prototype should focus on forecasting or anomaly detection rather than supervised health classification.

Regression models may be evaluated using MAE, RMSE, and R^2. Classification models may be evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices, as appropriate. Because environmental time series are seasonal, temporal validation and comparison with suitable baseline methods are recommended.

5. AI Agent and Decision Framework

Mangrove Guardian follows an observe–analyze–decide–act–report workflow. The first implementation can use modular, rule-based components, with later versions adopting an orchestration framework such as LangGraph or CrewAI. Potential modules include:

- Data Collection Agent: collects and validates new satellite observations.
- Prediction Agent: runs the trained machine-learning model.
- Decision Agent: combines predictions, thresholds, trends, and anomalies.
- Recommendation Agent: maps assessed risks to suitable conservation actions.
- Alert Agent: issues dashboard, email, or Telegram notifications.
- Reporting Agent: prepares periodic summaries and reports.

If vegetation conditions are stable and no significant anomaly is detected, the system assigns low risk. An NDVI decline below a seasonally appropriate threshold may trigger moderate risk. A sustained decline, especially when accompanied by an environmental anomaly or a model prediction of degradation, may trigger high risk and a recommendation for intensive monitoring or field verification. These prototype rules must be calibrated using domain expertise and validated observations.

6. Proposed System Architecture

The proposed architecture contains six logical layers: data acquisition, preprocessing and feature extraction, dataset management, machine learning, AI decision support, and visualization/reporting.

Layer 1 – Data Acquisition: Sentinel-2 and Sentinel-5P collections are queried for the Sundarbans area of interest and selected time period through GEE.

Layer 2 – Feature Extraction: Cloud-filtered Sentinel-2 surface-reflectance imagery is used to calculate NDVI. Sentinel-5P OFFL/L3 CO observations provide vertically integrated CO column density. Monthly or other regular composites create comparable time-series records.

Layer 3 – Dataset Management: Date, NDVI statistics, CO statistics, month, day of year, year, and optional trend features are stored in CSV files or a relational database.

Layer 4 – Prediction: A Random Forest model predicts either a continuous vegetation indicator or a categorical health/degradation class, depending on the labeling strategy.

Layer 5 – Decision Support: Model predictions are combined with thresholds and rates of change. For example, a sustained NDVI decline combined with elevated or increasing CO may increase the degradation-risk score.

Layer 6 – User Interface: A Streamlit dashboard displays current conditions, historical trends, predictions, risk status, alerts, and recommended actions.

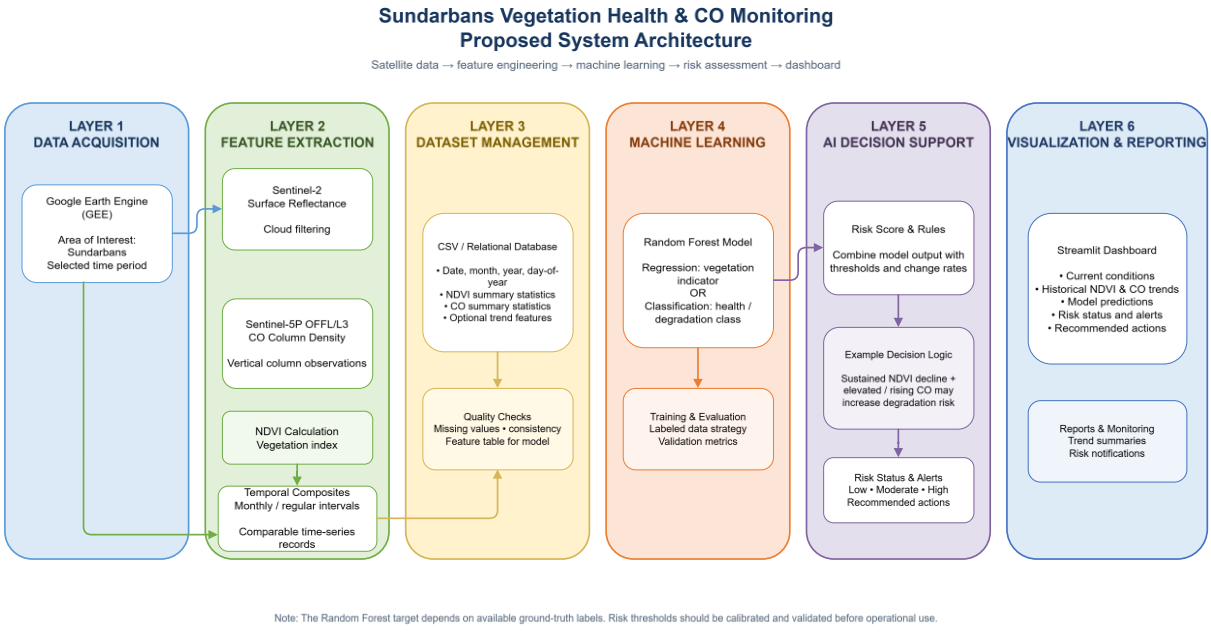


Fig. 1. Proposed system architecture for Sundarbans vegetation-health and CO monitoring.

7. Expected Results

The project is expected to generate (1) a repeatable NDVI time series for the Sundarbans coast and (2) a CO time series aligned with the NDVI observations. Planned outputs include environmental time-series plots, summaries of seasonal and long-term changes, a baseline machine-learning model, risk classifications, alerts, and an interactive dashboard. Potential scientific outcomes include identifying areas or periods of persistent vegetation loss and exploring relationships between vegetation anomalies and atmospheric conditions. Such relationships should be interpreted cautiously: correlation between CO and NDVI does not establish causation, and model uncertainty should be communicated clearly.

8. Future Enhancements

Future versions may incorporate rainfall, temperature, humidity, soil moisture, salinity, tidal conditions, cyclone tracks, water quality, and flooding. Additional vegetation indices, including EVI, NDWI, NDMI, and SAVI, could provide a more comprehensive ecosystem characterization. More advanced models, such as XGBoost, LSTM, ConvLSTM, U-Net, and vision transformers, could be compared with the Random Forest baseline. Explainable-AI methods such as SHAP may improve transparency. Sentinel-1 synthetic-aperture radar and drone imagery could complement optical monitoring during cloudy periods or at finer spatial scales. A conversational assistant could allow users to query ecosystem status in natural language, while a mobile application could enable field officers to submit geotagged observations. Other opportunities include carbon-stock estimation, biodiversity and habitat monitoring, restoration prioritization, and extending the approach to other mangrove ecosystems in India.

9. Conclusion

Mangrove Guardian proposes a research framework that integrates satellite remote sensing, machine learning, and AI-assisted decision support for proactive mangrove conservation. Sentinel-2 provides vegetation-condition information through NDVI, while Sentinel-5P contributes atmospheric context through CO observations. Google Earth Engine supports scalable geospatial processing, and a Random Forest baseline provides a starting point for prediction. A decision layer converts observations and model outputs into risk levels, alerts, and recommended actions. The key contribution is the end-to-end integration of these components in a modular architecture that can grow from a college-scale prototype into a broader monitoring platform. Scientific validity will depend on a well-defined AOI, careful sensor quality control, seasonal alignment, defensible labels, ground-truth validation, and transparent reporting of uncertainty. With these safeguards, Mangrove Guardian could support conservation decision-making in the Sundarbans and, eventually, other mangrove ecosystems.

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